GATE

Computer Science & Information Technology

Computer Networks

DPP: 1

ISO-OSI Stack

Q1 In the following pairs of OSI protocol layer/sub-layer and its functionality, the CORRECT pair is
(A) Network layer and Routing
(B) Data Link Layer and Bit synchronization
(C) Transport layer and End-to-end process communication
(D) Medium Access Control sub-layer and Channel sharing.

Q2 In OSI protocol layer and their functionality, which one is the CORRECT pair / s?
(A) Segmentation and reassembly are done at hosts only, not at each intermediate router.
(B) Application Layer allows the user or application to access the network.
(C) Session Layer establishes, maintains, and synchronizes the interaction between the communicating systems.
(D) Presentation Layer allows the process to add checkpoints or synchronization points in the stream of data.

Q3 In OSI layers and its functionality, which one is the CORRECT pair /s?
(A) Physical Layer maintains the same bit rate at the source and destination.
(B) Physical Layer synchronizes sender and receiver clocks.
(C) Physical layer is responsible for moving individual bits from one (node) to the next.
(D) Physical layer defines the type of encoding

Q4 Which of the following is/are CORRECT?
(A) Physical addresses in the packet are modified from HOP to HOP.
(B) Network Layer ensures packets belonging to the same message arrive intact and in order.
(C) Network layer uses physical addresses for addressing.
(D) Transport Layer treats every packet individually.

Q5 Match the following to one or more layers of the OSI model:

	List-I		List-II
1.	Transport Layer	a	Reliable process-to-process message delivery
2.	Data Link Layer	b	Route selection
3.	Network Layer	c	Defines frames
4.	Application Layer	d	Provides user services such as e-mail and file transfer
		e	Uses service point addressing
		f	Delivery of individual packets of data
		g	Delivery of each packet independently

(A) 1-g, 2-c, 3-b, 4-d

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(B) 1-b, 2-c, 3-a, 4-d
(C) 1-a, 2-c, 3-b, 4-d
(D) 1-e, 2-c, 3-f, 4-d

Q6 Which of the following statements is/are correct?
I. In the virtual circuit network, packets are never delivered out of order and In a datagram network, packets may be delivered out of order.
II. Virtual circuit network provides connection-oriented service, and Datagram network provides connectionless service.
III. Virtual circuit network is Highly reliable, and Datagram network is not reliable
IV. Virtual circuit networks are costly, and datagram networks are cost-efficient
(A) Only I (B) I & II
(C) II & III (D) All of these

Q7 Repeater functions at_____.
(A) Physical Layer
(B) Data Link Layer
(C) Network Layer
(D) Both (A) and (B)

Q8 Which of the following statements is/are TRUE according to TCP/IP stack
(A) IP hides the underlying network hardware from the network applications
(B) A set of interconnected physical networks that limit the range of an IP packet is called an "internet".
(C) A node which is connected to n links needs 1 physical-layer protocol.
(D) TCP/IP does not define any specific protocol for the Network Layer.

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Answer Key

Q1	(A, C, D)	Q5	(C, D)
Q2	(A, B, C)	Q6	(D)
Q3	(A, B, C, D)	Q7	(A)
Q4	(A)	Q8	(A, B)

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Hints & Solutions

Q1 Text Solution:
Option b is incorrect because bit synchronization is the functionality of physical layer.

Q2 Text Solution:
True. Segmentation and reassembly are done at hosts only, not at each intermediate router.
True. Application Layer allows the user or application to access the network.
True. Session Layer establishes, maintains, and synchronizes the interaction between the communicating systems.
False. Presentation Layer allows the process to add checkpoints or synchronization points in the stream of data.

Q3 Text Solution:
Physical Layer maintains the same bit rate at the source and destination.
Physical Layer synchronizes sender and receiver clocks.
Physical layer is responsible for moving individual bits from one (node) to the next.
Physical layer defines the type of encoding.
All True,

Q4 Text Solution:
A. True, MAC addresses of the next HOP are updated at every intermediate router.
B. False. At Network Layer, all the packets belonging to the same message are treated as different packets.
C. False, Network layer uses logical addresses for addressing.
D. False, Transport Layer ensures that the whole message arrives intact and in order

Q5 Text Solution:
Transport Layer:
1. Reliable process-to-process message delivery
2. Uses service point addressing called port numbers
Data Link Layer:
1. Defines frames
Network Layer:
1. Route selection
2. Delivery of each packet independent of the entire message
Application Layer:
1. Provides user services such as e-mail and file transfer

Q6 Text Solution:
All the statements are correct

Q7 Text Solution:
Repeater works at the Physical Layer.

Q8 Text Solution:
A. True: IP at the Network layer in TCP/IP to ensure Physical Network Independence, i.e. if you invent a new physical network, you can put it into service by implementing a new driver connecting network applications and the network.
B. True: it is the conceptual definition of the internet.
C. False: A node which is connected to n links needs n physical-layer protocols. The reason is that different links may use different physical-layer protocols.
D. False: TCP/IP defined a specific protocol at the Network layer called IP (Internet

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